

JONATHAN KOBRUS

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PROFILE

Reading MSc Bioinformatics at Birkbeck, University of London, with a year of industrial experience analysing large-scale whole genome sequencing datasets, developing bioinformatics pipelines, and supporting genomic product development at Circagene. Proficient in UNIX, Python, R, and genomics tools including GATK, Samtools, and Kraken2. Comfortable working independently in fast-paced environments and communicating technical findings to multidisciplinary teams. International upbringing fostering adaptability, cultural awareness, and effective collaboration within diverse teams.

SKILLS & EXPERTISE

Programming and tools

- Python, Bash, R, Java, Perl, SQL
- GATK, Samtools, BWA, QIIME2, Kaiju
- Docker, GitHub, Linux HPC environments
- BAM, FASTA, FASTQ, BED, JSON file handling

Bioinformatics & Genomics

- **Whole genome sequencing analysis** — Processed and interpreted human WGS data across pharmacogenetic, disease risk, and oral health applications at CircaGene.
- **Comparative & microbial genomics** — Processed 16S rRNA amplicon data for bacterial community analysis on an oral health project using QIIME2 and R; comparative genomic methods from MSc coursework.
- **Statistical & ML analysis** — Applied LDA for microbiome classification; statistical modules at MSc level covering hypothesis testing and data science methods.
- **Database management** — Automated genomic data workflows using Python and MongoDB; SQL database work using SQLite from MSc coursework.

Professional Skills

- **Scientific communication** — Presented technical findings fortnightly to mixed audiences at CircaGene; delivered staff-wide presentations at St Mary's.
- **Training & mentoring** — Restructured and supervised projects for 15 interns; taught Python programming to adult learners via Code First Girls.
- **Working to deadlines** — Managed concurrent bioinformatics and operational responsibilities across start-up and high-volume coordination roles simultaneously.

EDUCATION

Birkbeck University of London

2025 - present

MSc Bioinformatics

- **Thesis** — *"Predicting protein NMR parameters from structure"*
- **Modules** — Molecular Basis of Life, Biocomputing, Statistics and Data Science, Sequence and Omics.

Brunel University London

2019 - 2023

2:1 BSc Life Sciences - Major: Computer Science - Minor: Biochemistry

- **Thesis** — *"The Role of Inhibitory Control in Periodontitis and the Mediating Role of Diet"* Used Network Mendelian Randomisation to find causal genetic factors between inhibitory control (IC) and gum disease (GD), and a sugar-free diet. Used data from the UK Biobank and mined data from publicly available data.
- **Relevant modules** — Statistics -omics and Bioinformatics (**MSc level module**), Pharmacology, Logic and Computation, Algorithms and their applications, Analytical Biochemistry, Quantitative research.
- **Industrial Placement** — Internship at CircaGene in the Bioinformatics and R&D department under the supervision of Iandry Rabearivelo and Francois Pailler.

Lincoln Minster School

2016 - 2018

A Levels

- Biology, Chemistry, and Computer Science.

PROFESSIONAL EXPERIENCE

CircaGene

Bioinformatics Intern

2021 - 2022

London

- Analysed large-scale WGS datasets from customer saliva samples using GATK, Samtools, BWA, Perl, and Bash, enabling generation of genomic health reports covering pharmacogenetics, oral health, and disease risk profiling.
- Investigated antimicrobial resistance (AMR) detection methodologies and developed a read-based AMR analysis pipeline for whole genome sequencing (WGS) data using reference-alignment approaches to analyse periodontal pathogens including *Porphyromonas gingivalis* and *Treponema denticola*, identifying antimicrobial resistance markers such as erythromycin esterase genes.
- Applied statistical and machine learning approaches, including Linear Discriminant Analysis (LDA), to classify oral microbiome profiles as healthy or periodontitis-associated after independently researching alternative analytical methods and consulting statistical experts to resolve failed implementations.
- Developed and optimised bespoke bioinformatics pipelines for genomic analysis projects, including pharmacogenetic screening and Alzheimer's-associated variant identification, improving report functionality and expanding disease coverage for customers.
- Designed and implemented scripts using APIs and BED file generation workflows to automate mutation annotation and genomic coordinate extraction for multiple analysis panels, improving development efficiency for genomic products.
- Mentored and supported approximately 15 interns by redesigning projects into achievable deliverables aligned to company objectives, improving project completion rates and reducing management burden on senior colleagues.

Code First Girls

Python Instructor

Oct 2021-Dec 2021

London

- Taught structured introductory Python programming courses to adult learners in collaboration with a team of instructors, improving accessibility of technical concepts for non-specialist audiences.
- Adapted teaching to meet the needs of adult and part-time learners, ensuring lessons were accessible and engaging regardless of prior experience.
- Demonstrated core programming concepts step-by-step by live-coding examples, explaining syntax and logic clearly, and offering one-to-one guidance during practical tasks.

St Mary's London

Co-ordinator

2023 - 2025

London

- Ensuring smooth running of events, assisting in the general running of the office, coordinating volunteer rotas, maintaining and updating the company website, overseeing office IT, and point of contact for the organisation to the public.
- Onboarded and trained new staff members, explaining departmental processes and ensuring consistency in administrative duties.
- Improved admin processes by simplifying expense and receipt submissions, increasing staff compliance from under 20% to over 80%, easing pressure on finance staff.
- Recognised by senior leadership for setting and maintaining exceptionally high documentation standards, enabling "one of the best administrations to date" in a fast-paced, zero-tolerance environment.
- Delivered confident presentations to the entire staff team, explaining key processes (e.g. recording credit card expenses), and provided pre-event briefings to volunteer teams outlining roles and responsibilities.

Tesco

Customer service assistant

2026

Whitehaven

- Delivered high-quality customer service in a fast-paced retail environment, recognised informally by colleagues for approachability and professionalism within the first month.
- Handled customer queries and transactions accurately at the service desk and tills, demonstrating reliability and attention to detail in a customer-facing role.